Unit 18 - Variance Estimation and Improved Sensitivity: Pitfalls and solutions

(1) What does "Common Pitfalls" mean in this context?

→ It refers to frequent mistake practitioners make when estimating variance on online experiments (A/B testing), which can mislead experiment conclusions.

key Pitfalls (with Analogies, Math, and Code).

☑ Pitfall 1: Using Raw click-Through Rate (CTR) Variance Instead of Per-User Metrics.

#Concept

When you analyze variance using raw CTR (total clicks/total impressions), you underestimate the true variability. This is especially bad when user behavior is skewed (some click 0, others click 100 times)

Real-world Analogy

Imagine two websites:

- Site A: users, each clicks once → CTR = 1 click/user
- Site B: 10 users click 10 times, 90 click 0 times
  → CTR still = 1 click/user.

Both have the same CTR, but site B has much more variability, which is invisible in the aggregate CTR

↳ What's Going on?

we are comparing two experiments/sites that both report the same average CTR, ~~but~~ but the distribution of user behavior is very different

Let's define:

- CTR (Click-Through Rate) = Total Clicks / Total Users

This is a point estimate — a single number. But it hides the distribution behind that number

☑ Site A : Uniform Users

- 100 users
- Every user clicks once

Total Clicks = 100

$$CTR = \frac{100 \text{ clicks}}{100 \text{ users}} = 1$$

- Each user's click count

[1, 1, 1, ... 1] (100 times)

- Variance of user clicks

Var (X) = 0

(because everyone behaves the same)

⚠ Site B : skewed Behavior

- 100 users total
- 10 users clicked 10 times each = 100 clicks
- 90 users clicked 0 times

Total clicks = 100

$$CTR = \frac{100}{100} = 1$$

- Each user's click count:

[10, 10 ..., 10 (10 times), 0, 0, ... 0 (90 times)]

- Variance

Var (X) > 0

(there is huge spread: some are 10, some are 0)

2. Why This is a pitfall in A/B Testing?

When you run an experiment and look at just the average CTR, you might think:

- "Great! CTR is 1 for both variants — experiments is stable"

But:

- Site A has stable user behavior - confidence interval around mean is tight.
- Site B has highly variable behavior → CI is wide due to high variance.

⇒ if you don't account for variance, your statistical test will be misleading

You might:

- Overestimate statistical confidence
- Misjudge significance
- Draw wrong conclusion about treatment/control difference.

Math Behind It:

Let:

- X_i : Clicks by user i
- N : Number of users
- Mean: $\bar{x} = \frac{1}{N} \sum_{i=1}^{N} X_i$
- Correct Variance (per users):

$$Var(\bar{X}) = \frac{1}{N} \cdot Var(X_i)$$

But if you use:

$$\text{Var}\left(\frac{\text{Total clicks}}{\text{Total Impressions}}\right)$$

It's not reflecting user-level behavior, just aggregate behavior → misleading.

[V] Pitfall 2: Not Accounting for Correlation within Units (e.g. Users or sessions)

Concept

You assume observations are independent, but they're not! e.g. Same user clicks multiple → correlated behavior → underestimated variance.

Real-World

You're measuring how happy customers are. If one customer talks 10x, you can't treat it like 10 independent users. They may be just one angry person!

So you must cluster by users.

Math

If you ignore intra-user correlation, you assume

$$\text{Var}(\text{Total}) = \sum \text{Var}(X_i)$$

But in reality

$$\text{Var}(\text{Total}) = \sum \text{Var}(X_i) + \sum \text{Cov}(X_i, X_j)$$

1) Real-world Example: E-commerce website - "Buy Now" Button Experiment.

You're running an A/B test on your website.

→ Goal: see if a new "Buy Now" button design (Treatment)

→ Metric: Number of purchases per session.

→ Each user visits the website multiple

⇒ Experimental Setup:

Let's say you have:

- N = 2 users in each group (for simplicity)
- Each user has n_i = 2 sessions.

⇒ Control Group (A)

user	Session 1	Session 2
U1	1 purchase	0 purchase
U2	1 purchase	1 purchase

Total purchases = 1 + 0 + 1 + 1 = 3

⇒ Treatment Group (B)

User	Session 1	Session 2
U1	1 purchase	1 purchase
U2	1 purchase	1 purchase

Total purchases = 1 + 1 + 1 + 1 = 4

* what is session ? *

→ In the context of web analytics or A/B testing, a session refers to:

> A single visit by a user to a website or app, usually within a defined time window.

Suppose you're shopping online:

1. Monday: You visit Amazon, browse 10 items, then leave after 20 minutes
   → that's 1 session
2. Tuesday: You return, check 3 products, buy 1 item and log out.
   → that's another session.

! You (the user) now have 2 sessions.

→ Why is this important in A/B Testing?

Because:

- Many metrics (clicks, purchases, time-on-site) are collected per session
- But users, not sessions, are the independent units in experiments.
- So if you don't group metrics per user, you're treating repeated sessions like they came from different people (false independence)

X Case 1: Wrong Approach – Assume Sessions Are Independent.

Treat all 4 sessions in each group as independent observations

△ Control Group A

Observations: [1, 0, 1, 1]

- Mean: $\mu_A = \frac{1+0+1+1}{4} = \frac{3}{4} = 0.75$
- Variance:

$$Var_A = \frac{1}{3}[(1-0.75)^2 + (0-0.75)^2 + (1-0.75)^2 + (1-0.75)^2] = \frac{0.75}{3} = 0.25.$$

△ Treatment Group B

Observation [1, 1, 1, 1]

- Mean: $\mu_B = \frac{4}{4} = 1$
- Variance: $Var_B = 0$ (All values same)

Inference

- You estimate:
  - Control mean = 0.75, Var = 0.25
  - Treatment mean = 1.0, Var = 0
- So, treatment seems much better, and you might believe the difference is significant.

⚠ But this is wrong because sessions within the same users are correlated.

✓ Case 2 : Correct Approach - Group by User

Group session data per user, then compute user-level variance (i.e. assume users are independent)

⇒ Control GroupA Example

Step1 : Aggregate sessions for each user.

- U1 : total purchases = 1 + 0 = 1
- U2 : total purchases = 1 + 1 = 2

So now you have user-level data

User-level purchases : [1, 2]

Step2 : Calculate mean

$$\mu_A = \frac{1+2}{2} = \frac{3}{2} = 1.5$$

Step3 : Calculate variance between users

Formula for variance

$$Var = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2}{2 - 1}$$

Substitute values:

$$= \frac{(1-1.5)^2 + (2-1.5)^2}{1}$$

$$= \frac{0.25 + 0.25}{1} = 0.5.$$

✅ This is the true variance between users in the control group.

⇒ Treatment Group B Example

Aggregate per user:

- U1 : 1 + 1 = 2
- U2 : 1 + 1 = 2

User-level purchases = [2, 2]

Mean:

$$\mu_B = \frac{2+2}{2} = 2$$

Variance

$$Var_B = \frac{(2-2)^2 + (2-2)^2}{2-1} = \frac{0}{1} = 0.$$

✅ All users in Treatment bought the same amount → no variance.

Formula:

(1) Session-level:

Mean: $\bar{x}_{session} = \frac{1}{\sum_{i=1}^{N} n_i} \sum_{i=1}^{N} \sum_{j=1}^{n_i} x_{ij}$

This is just the total sum across all sessions, divided by the total number of sessions.

Variance: $Var_{session} = \frac{1}{\sum_{i=1}^{N} n_i - 1} \sum_{i=1}^{N} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_{session})^2$

⚠ Assumes all sessions are independent, which is often false – sessions from the same user are correlated.

(2) User-Level:

Mean: $\bar{x}_{user} = \frac{1}{N} \sum_{i=1}^{N} X_i = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij} \right)$

This reflect the average behavior per user, not per session.

Variance: $Var_{user} = \frac{1}{N-1} \sum_{i=1}^{N} (X_i - \bar{x}_{user})^2$

This is the correct estimator if users are the randomized unit (which is true in almost all A/B tests)

Why User-level is Better than Session-level Aggregation

Session-level (incorrect way):

- Treats each session (visit) as a new person
- In reality, it's the same user behaving in multiple sessions.
- Ignoring that gives false independence → underestimate variance

User-level (correct way):

- Groups all behavior by user

- Respects the fact that users are the unit of randomization
- Gives true estimate of variance across users

Pitfall 8: Using Wrong Unit of Analysis

Real-world Analogy:

Let's say you're testing ads:

- Ad shown 1000 times (impression)
- 100 users (some saw it 10x)

If you analyze on impression-level but randomized on user-level, the analysis is wrong. You're breaking the experiment design logic.

Math

If you randomized by users, analysis should be aggregated by user. Failing this leads to unit mismatch bias

$$\mathbb{E}[\hat{\Delta}_{\text{impression}}] \neq \mathbb{E}[\hat{\Delta}_{\text{user}}]$$

1) Scenario Setup: Ad Click-Through Rate (CTR) A/B Test

Goal: You want to test if ad design B get more clicks than ad design A.

Experiment design:

- Users are randomly assigned to either Control (A) Treatment (B)

* Final Insight

$$\mathbb{E}[\hat{\Delta}_{impression}] \neq \mathbb{E}[\hat{\Delta}_{user}]$$

Because:

- Impressions are not independent units
- Heavy users (many impressions) bias impression-level estimates.
- Your p-values, CI and significance will be wrong.

☑ Fix:

To correctly estimate treatment effect

1. Randomise by user
2. Aggregate metrics per user
3. Compute statistic (mean, variance, significance) per user.

** What is an Impression? **

In digital advertising and analytics:

> An impression is one views of an ad, post or webpage by a user.

Real-World Examples

* On Instagram:

→ You scroll past a sponsored post → that is 1 impression.

→ You scroll past the same ad again later → another impression

* On an E-commerce site:

- You visit a product page showing an ad → 1 impression
- You refresh the page → now it's 2 impressions
- You return tomorrow → now it's 3 total impressions.

Each time the ad is show to a user = 1 impression, regardless of clicks or engagement

How Impressions Relate to A/B Testing:

In an ad experiment

User	Group	Impressions	clicks
U1	Control	10	1
U2	Control	5	1

User	Group	Impressions	Clicks
U1	Control	10	1
U2	Control	5	1

- U1 saw the ad 10 times, clicked once → 10 impressions | 1 click
- U2 saw it 5 times, also clicked once.

But if you count each impression as an independent observation, you violate the fact that:

> The same user is being shown the same ad repeatedly, so these events are correlated.

Summary

Term	Meaning
User	A person visiting your site or app
Session	A continuous visit by a user (e.g. 30 mins)
Impressions	A single ad or content view by a user
Click	The user interacts with the impression.

☑ Pitfall 9: Overfitting to Variance Reduction Techniques.

Concepts:

If you apply techniques like covariate adjusting, bucketing or pre-period matching too aggressively, you risk overfitting to noise and hurting generalizability.

Real-world Analogy:

You adjust everyone's score using a fancy formula that works great on training data, but when a new batch comes in → poor performance.

Over-tuned variance reduction = low robustness.

⚠ Problem: what is Variance Reduction Overfitting?

⊙ Goal of variance reduction:

In A/B testing, we often apply techniques like:

- Covariate adjustment (e.g. CUPED)
- Pre-period matching
- Bucketing by historical activity

... to reduce the variance of estimated treatment effects so we get tighter confidence intervals and statistical significance faster.

⚠ Pit fall:

If these adjustments are too aggressive or tuned on the experiment itself, they can overfit noise (just like in ML), making results look good on this experiment but fail to generalize to future users or contexts

Math Examples: Overfitting in Variance Reduction

⊙ Goal: You run an A/B test on your e-commerce site to test a new "Buy Now" button design (Treatment B vs Control A).

⚠ The Mistake:

You use a covariate (e.g. last week's time spent on site) to adjust your experiment outcome. But you tune the adjustment too much — making the result look better than it is (classic overfitting).

1	2
3	4

Math setup.

We simulate:

User	Group	Pharmalnia (X) = Time on site last week.	Purchases this week (Y)
U1	Control	5 minutes	10
U2	Control	8 minutes	12
U3	Treatment	6 minutes	11
U4	Treatment	7 minutes	12

Let's walk through this.

☑ Setup 1: No adjustment.

Group means of Y:

- Control: $\bar{Y}_A = \frac{10+12}{2} = 11$
- Treatment: $\bar{Y}_B = \frac{11+12}{2} = 11.5$

→ Estimated difference

$$\hat{T} = 11.5 - 11 = 0.5$$

So it looks like Treatment B is better, but let's apply covariate adjustment.

☑ Step 2 : Adjusting for Pre-metric (X)

We use this formula

$$y_i^{adj} = y_i - \beta(x_i - \bar{x})$$

where :

- y_i : observed purchases
- x_i : pre-experiment behavior
- $\bar{x}$: average x
- β : weight (we'll pick one)

Step 2.1 : calculate $\bar{x}$

$$\bar{x} = \frac{5+8+6+7}{4} = 6.5.$$

Step 2.2 : Try $\beta = 1$ (arbitrary)

Now compute

User	Group	x	y	$y^{adj} = y - (x - 6.5)$
U1	A	5	10	$10 - (5 - 6.5) = 11.5$
U2	A	8	12	$12 - (8 - 6.5) = 10.5$
U3	B	6	11	$11 - (6 - 6.5) = 11.5$
U4	B	7	12	$12 - (7 - 6.5) = 11.5$.

Adjusted Group Means

- Control : $(11.6 + 10.5)/2 = 11$
- Treatment : $(11.6 + 11.2)/2 = 11.5$

→ Adjusted $\hat{\tau} = 0.5$ → Same result

But now suppose we tune β further to minimize variance within group — and we do this after seeing the results.

That would be overfitting. You're using the test data to adjust the model — this is like data leakage in ML.

⚠ What Overfitting to Variance Reduction looks like

Let's pretend we fit a model:

$$Y_i = \alpha + \tau T_i + \beta X_i + \varepsilon_i$$

But you pick β to reduce the p-value (make the result significant) instead of following a fixed plan.

Result:

- Smaller variance
- Smaller p-value
- False positive

⇒ Final Insight:

> Variance reduction (like covariate adjustment) is good — but if you tune it on the outcome, you're cheating.

It's like measuring someone's height after they've jumped — the measurement is no longer stable.

⇒ Summary

Step	Good?	Why
Adjust using pre-experiment data	✓	Reduces variance if pre-planned.
Tune adjustment after seeing Y	✗	Overfits, leads to biased treatment effects.

② Improving Sensitivity 196

X ① What is Improving Sensitivity? X

→ It refers to method that decrease variance in your experiment, making it easier to detect smaller treatment effects (i.e. increasing statistical power)

Why This Matters

A/B test often suffer from high variability - metrics like revenue or engagement can be noisy. Without reducing this noise, you'd need huge samples to find real effects. Improving sensitivity helps you:

- use fewer users or time
- Detect smaller improvements
- Reduce false negatives

X ② Key Techniques X

① Covariate Adjustment (CUPED)

- uses pre-experiment data (e.g. past purchase behavior) to reduce variance
- Adjusted metric

$$Y_i^{adj} = Y_i - \beta(X_i - \bar{X})$$

where X_i is a baseline covariate, and β is estimated from regression

- Effectively removes predictable variation, cutting standard error.

② Triggering

- Only includes users who meet a certain behavioral condition (e.g. saw an ad)
- Results in a more relevant user base, focusing on people affected by the treatment.
- Helps remove irrelevant noise from users who never encounter the treatment.

③ Bucketing (Stratification)

- Divide user into groups (e.g. by location or past behavior).
- Analyze within buckets and pool results.
- Ensures better comparisons between similar users.

X ④ Math: Variance with Covariate Adjustment X

without adjustment:

$$Var(y_i) = \sigma_y^2$$

with adjustment using X:

$$y_i^{adj} = y_i - \beta(x_i - \bar{x})$$

where β is chosen so:

$$\beta = \frac{Cov(x, y)}{Var(x)}$$

Then adjusted variance becomes:

$$Var\left(\bar{Y}_1^{adj}\right) = \sigma_y^2 (1 - R^2).$$

- $R^2 = [Corr(X, Y)]^2$
- If $R^2 = 0.4$, variance is reduced by 40%
- This boosts sensitivity by $\sqrt{1/0.6} \approx 1.29$

✗ (5) Connections to Other Topics ✗

- Common Pitfalls (p. 193): Variance reduction only works if done properly, without overfitting or using post-treatment data.
- Variance of Other Metrics (p. 198). Technique apply not just to means - also to ratios, rates, percentiles etc.
- Triggering comes up again in unit 20 (p. 208), with deeper math on optimal user selection.

✗ (6) Bottom Line ✗

Improving sensitivity means reducing variance smartly, not just increasing sample size. Key methods include:

- CUPED: leverage prior data
- Triggering: Focuses on relevant users
- Bucketing/stratifications: balances groups

Applied correctly, these techniques let you detect real effects faster and more reliably.

(3) Variance of other Statistics 198

Variance of other statistic emphasing cokg variance multans for non-averaage metrics, with math analogies and a practical Python demo.

X (1) Why This Matters? X

Most discussions focus on variance of means, but real-world experiments often involve other metrics:

- Ratios (eg CTR = clicks / impressions)
- Rates (e.g. conversion rate)
- Percentiles (e.g. 90th percentile of latency)
- Ratios of sums, or derived

These require different variance estimation approaches, because the standard formula assumes means of i.i.d. data.

X (2) Key Concepts. X

1. Delta Method for Ratios.

Use when you're dealing with $R = \frac{Y}{X}$ (e.g. CTR). Variance can be approximated as

$$Var(R) \approx \frac{1}{\bar{X}^2} Var(Y) + \frac{\bar{Y}^2}{\bar{X}^2} Var(X) - 2\frac{\bar{Y}}{\bar{X}^3} Cov(X,Y)$$

This is the Delta Method. It yields valid confidence intervals for ratios like CTR or conversion rate.

Problem Setup: Real-World Example

Scenario:

You're testing a new ad design. You run A/B test.

For each user, you record:

- x_i: Number of impressions (how many times the ad was shown to user i)
- y_i: Number of clicks (how many times the user clicked)

The metric of interest is:

$$R = \frac{\sum_i y_i}{\sum_i x_i}$$

This is the CTR — the total number of clicks divided by the total number of impressions.

Now, you want to calculate $Var(R)$ — i.e. how much uncertainty is there in your CTR estimate?

Why We Can't Just Use $Var(y)/n$

CTR is ratio of two random variables.

- $y = \sum_i y_i$ is random
- $x = \sum_i x_i$ is also random

So, we cannot use the usual variance formula for means:

$$Var(\bar{y}) = \frac{\sigma^2}{n}$$

We need something more powerful: the Delta Method.

Delta Method: what it does

The Delta Method is used to approximate the variance of a function of a random variable.

Let:

- $R = f(X, Y) = Y/X$

Then the 1st-order Taylor approximation of the variance is:

$$Var(f(X,Y)) \approx \left(\frac{\partial f}{\partial X}\right)^2 Var(X) + \left(\frac{\partial f}{\partial Y}\right)^2 Var(Y) + 2 \frac{\partial f}{\partial X} \frac{\partial f}{\partial Y} Cov(X,Y)$$

⇒ Apply to CTR

Let $f = \frac{Y}{X}$. Compute partial derivatives

- $\frac{\partial f}{\partial Y} = \frac{1}{X}$
- $\frac{\partial f}{\partial X} = -\frac{Y}{X^2}$

So the variance becomes

$$Var(R) \approx \frac{1}{\bar{X}^2} Var(Y) + \frac{\bar{Y}^2}{\bar{X}^4} Var(X) - 2 \cdot \frac{\bar{Y}}{\bar{X}^3} Cov(X,Y)$$

This gives you a good estimate of the uncertainty in CTR, even though it's a ratio of sums.

☑ Real Data Example (by hand)

Let's take data from 4 users

User	Impressions (x)	Clicks (y)
U1	10	2
U2	8	1
U3	12	3
U4	6	0

Compute sample means:

$$\bar{x} = \frac{10+8+12+6}{4} = 9$$

$$\bar{y} = \frac{2+1+3+0}{4} = 1.5.$$

Sample variances:

$$\text{Var}(x) = \frac{(10-9)^2 + (8-9)^2 + (12-9)^2 + (6-9)^2}{3}$$

$$= \frac{1+1+9+9}{3} = \frac{20}{3} \approx 6.67$$

$$\text{Var}(y) = \frac{(2-1.5)^2 + (1-1.5)^2 + (3-1.5)^2 + (0-1.5)^2}{3}$$

$$= \frac{0.25 + 0.25 + 2.25 + 2.25}{3} = \frac{5}{3} = 1.67.$$

Covariance:

$$\text{Cov}(x,y) = \frac{1}{3}\sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{3}[(1)(0.5) + (-1)(-0.5) + (3)(1.5) + (-3)(-1.5)]$$

$$= \frac{1}{3}[0.5 + 0.5 + 4.5 + 4.5] = \frac{10}{3} \approx 3.33$$

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plug into Delta formula

$$Var(R) \approx \frac{1}{g^2} \cdot 1.67 + \frac{(1.5)^2}{g^4} \cdot 6.67 - 2 \cdot \frac{1.5}{g^3} \cdot 3.33$$

$$Var(R) \approx 0.0092$$

standard error $\approx \sqrt{0.0092} \approx 0.096$.

Interpretation

- Estimated CTR: $\frac{\sum y}{\sum x} = \frac{6}{36} = 0.1667$.
- Standard Error (SE) $\sim$ 0.096.
- 95% CI

$$0.1667 \pm 1.96 \cdot 0.096 \approx [-0.02, 0.33]$$

This shows CTR estimate has high uncertainty with such a small sample - you can't trust the estimate without this correct variance treatment

2. Ratio of Averages vs. Average of Ratios

- Naive: compute CTR per user (clicks / impr), then take average.
- Normalized: Divide sum of clicks by sum of impressions across users.

These can differ significantly, especially with heterogeneous user behavior, leading to Simpson's paradox.
Always choose the variant with lower variance and correct bias.

1. Ratio of Averages (Normalized CTR)

$$CTR_{normalized} = \frac{\sum_i clicks_i}{\sum_i impressions_i}$$

2. Average of Ratios (Naive Per-User CTR)

$$CTR_{naive} = \frac{1}{n} \sum_i \left(\frac{clicks_i}{impressions_i} \right)$$

Why They're Not the Same (unless every user has same impressions)

The Normalized CTR weights each user proportionally to their impressions.
The Naive CTR gives equal weight to every user, even those who had just 1 impression.

This can lead to very different outcomes if:
→ User impressions vary wildly
→ Click behavior differs across user size buckets.

Real-World Example: Influencer vs Regular User

User	Impressions	Clicks	CTR (User)
A (influencer)	1,000	100	0.10
B (small User)	10	3	0.30

⚠ Naive CTR (Average of Ratios)

$$CTR_{naive} = \frac{0.10 + 0.30}{2} = 0.20$$

Misleading ! suggests a 20% CTR. But this overweights the small user.

☑ Normalized CTR (Ratio of Averages):

$$CTR_{normalized} = \frac{100+3}{1000+10} = \frac{103}{1010} \approx 0.102$$

Much closer to the true weighted behavior, as the influencer dominates the impressions.

Pure Math Generalization.

Let:

- x_i = impressions$_i$
- y_i = clicks$_i$
- $r_i = y_i / x_i$ = per-user CTR

Then:

- Average of Ratios (Naive):

$$\frac{1}{n} \sum_{i=1}^{n} \frac{y_i}{x_i}$$

Each user get equal weight, even if they had only 1 impression.

- Ratio of Averages (Normalized):

$$\frac{\sum y_i}{\sum x_i} = \frac{\sum x_i r_i}{\sum x_i}$$

This is a weighted average of the per-user CTR, weighted by impression count x_i.

⚠ Simpson's Paradox in Action.

If high-CTR user also have low impressions, and vice versa, the naive average gets skewed upward. The normalized CTR avoids this bias.

- Ads (CTR)
- E-commerce (conversion rates)
- Engagement (watch-through, click depth).

3. Weighted Averages with Correlation Adjustment.

When users have different sample sizes and correlation in repeated measures, the optimal weights are:

$$w_i \propto \frac{n_i}{1 + (n_i - 1)\rho}$$

This gives the UMVUE (optimal variance) estimator,

where,

- n_i = number of repeated measurement (e.g. sessions) for user i
- ρ = intra-user correlation (how similar user's repeated values are)
- w_i = weight for user i

Why this formula?

Let's break it down:

- if $\rho \approx 0$: all visits are independent → then $w_i \propto n_i$ (more data = more trust)

- If $\rho = 1$: all visits are identical → then $w_i = 1$
  (one value gives full info, adding more gives nothing new)

- If for $0 < \rho < 1$: diminishing returns for repeated data.

> This is a statistical shrinkage - accounts for ~~that~~ the fact that users with 1000 correlated visits don't give 1000× more info than a user with 1 visit.

Scenario:
We want estimate the overall revenue per session, using from 3 users. Each user has multiple sessions, and their session revenue may be correlated.

Data:

User	n_i (sessions)	$\bar{x}_i$ (average revenue / session)
A	10	2.00
B	1	2.50
C	50	1.80

Let's say intra-user correlation $\rho = 0.2$.

* Intra-User Correlation *

→ Intra-user correlation — often written as ρ (rho) — measures how similar a user's behavior is across their multiple sessions.

Suppose a user visits your site 10 times:

- If their behavior is highly consistent (e.g. always spends $5, always clicks) ⇒ then p is close to 1 ⇒ p is high ⇒ treat actions as less informative.
- If their behavior is random across visits (clicks something, not others) ⇒ p is closer to 0 ⇒ p is low ⇒ treat ~~act~~ repeated actions as nearly independent.

Continue on Example ---

Objective: find the best unbiased estimate of the overall session-level mean using:

$$\theta = \sum_{i=1}^{n} w_i \bar{x}_i$$

where:

- $w_i \propto \frac{n_i}{1 + (n_i - 1)\rho}$
- And $\sum w_i = 1$ (normalize to sum to 1)

Step-by-step Math:

Step 1: Compute Unnormalized Weights

Apply:

$$w_i' = \frac{n_i}{1 + (n_i - 1)\rho}$$

For User A:

$$w'_A = \frac{10}{1 + (10-1) \cdot 0.2} = \frac{10}{1+1.8} = \frac{10}{2.8} \approx 3.571$$

For User B:

$$w'_B = \frac{1}{1 + (1-1) \cdot 0.2} = \frac{1}{1} = 1.$$

For User C:

$$w'_C = \frac{50}{1 + (50-1) \cdot 0.2} = \frac{50}{1 + 9.8} = \frac{50}{10.8} = 4.630.$$

Step 2: Normalize Weights

Compute total:

$$\text{Total} = w'_A + w'_B + w'_C = 3.771 + 1 + 4.630 = 9.201.$$

Now normalize each:

User		Normalized Weight w_i
A	$\frac{3.571}{9.201}$	≈ 0.388
B	$\frac{1}{9.201}$	≈ 0.109
C	$\frac{4.630}{9.201}$	≈ 0.503.

Step 3: Compute Final Weighted Average

$$\hat{y} = 0.388 \cdot 2.00 + 0.109 \cdot 2.50 + 0.503 \cdot 1.80.$$

$$\hat{y} = 0.776 + 0.273 + 0.905 = \boxed{1.954}.$$

Step 4: Compare with Naive Average

$$\text{Naive Mean} = \frac{2.00 + 2.50 + 1.80}{3} = \frac{6.30}{3} = 2.10.$$

Final Insight:

Method	Estimate
Naive Average	2.10
UMVE Weighted Avg	1.954

→ The naive average overweights the single-session session users (13).

→ The UMVUE corrects for both sample size and correlation, giving us a statistically optimal estimate.

4) Connections.

- Common Pitfalls (p. 193): Don't treat these statistics with mean-based formulas - variance gets underestimated
- Sensitivity (p. 196): After variance is computed correctly, techniques like covariate adjustment or triggered samples can further reduce it.
- Triggering & stratification: Ratio metrics often benefit from carefully segmenting and weighting users.

5) Bottom Line

1. Identify your statistics (CTR, ratio, percentile, etc)
2. Choose the right variance method (e.g., Delta Method, weighted estimator)
3. Avoid naive formulas that assume i.i.d.
4. Use proper weighting to account for sample size and correlation.

This ensures your CI and p-values are trustworthy — not misleading.